

Course “Object-Oriented Programming in C++”**Week №9*****Homework*****Exercise 1.**

Implement the database of the traffic police based on the traffic fines using a binary tree. the car number will be a key, and the list of offences will be a node value. If the receipt is added for the first time, a new node will appear on a binary tree, and the data on the offense will appear in the list; if the receipt is added not for the first time, the data are entered into the existing list. the following operations should also be implemented:

- Print the full database (by car numbers and list of their offenses)
- Print the data by the specified number
- Print the data by the range of numbers
- Save and load data from the file.

Exercise 2.

Two text files are given. Find out whether their strings match. If they don't match, then output from each file a string which doesn't match.

Exercise 3.

Text file is given. You should create a new file and record to it the following statistics according to the source file:

- Number of characters
- Number of strings

- Number of vowels
- Number of consonants
- Number of digits.

Exercise 4.

Caesar cipher is one of the simplest and most widely known encryption techniques. It is a type of substitution cipher in which each letter in the plaintext is replaced by a letter some fixed number of positions down the alphabet.

Example

Encryption using key 3

Plaintext: THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG

Ciphertext: QEB NRFZH YOLTK CLU GRJMP LSBO QEB IXWV ALD

Text file is given. Encrypt it using Caesar cipher. Results must be recorded to another file.